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Effectiveness of Therapeutic Oriented Software in Mental Health Treatment

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Abstract

Therapeutic-oriented software is a new field that is still mostly unknown to the general public. With the continuing rise of mental health conditions and the need for additional and alternative ways of assisting with the treatment of individuals, digital mental health technologies are being investigated as alternatives for assisting with the treatment of people with mental health conditions. Deaths attributed to mental health conditions have been on the rise, and as such, the effectiveness research approach is utilized to find and measure how effective digital therapeutic software truly is in treating or assisting with the treatment of people with mental health conditions. Artificial Intelligent (AI) chatbots, digital mental health interventions, affective computing, and mental health-oriented software are investigated. Costs, privacy, security, ethicality, and morality are taken into consideration throughout to determine some of the costs and major concerns or possible implications of using therapeutic oriented software in treating people with mental health conditions. Key findings include the true effectiveness of therapeutic oriented software, considerations before and during the development of therapeutic software, development processes during the development of therapeutic software, the maintenance of therapeutic software, and hardware limitations.

Keywords: Mental Health, Artificial Intelligence, Therapeutic Oriented Software, Affective Computing, Digital Mental Health Interventions, Positive Computing, Chatbots, Privacy, Security, Ethical, Moral.

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Effectiveness of Therapeutic Oriented Software in Mental Health Treatment

Introduction

The effectiveness of therapeutic-oriented software in successfully assisting with the treatment of individuals with mental health needs is an area with limited focus and clarity (Sundaram et al., 2024). There are various approaches to analyzing the costs of creating such software in exchange for achieving desired results that it can be difficult to measure true effectiveness (Lehtimaki et al., 2020). The primary considerations taken into account when developing, implementing, and deploying therapeutic software come in the forms of costs which are primarily monetary focused, privacy which consists of data collection transparency, security in the form of potential data leaks, ethicality in terms of potentially replacing human jobs, and morality in terms of the potential for improper treatment of users (Wu et al., 2024). These considerations are used imperatively to help determine the true effectiveness these software applications can have with assisting in the treatment of individuals with mental health needs when measuring some of the key resources it takes to create therapeutic software with respect to successful results.

Historically, the number of cases of individuals with mental health needs has been increasing, with psychosis, depression, and anxiety being the most prominent in mental health related deaths (Walker et al., 2025). More specifically, Post Traumatic Stress Disorder (PTSD) is a prevalent mental health related issue faced in the United Statutes, with approximately 6 percent of adults ultimately dealing with or facing PTSD at some point in their life (Wu et al., 2024). Untreated mental health conditions leave significant room for danger to a person, and as such, it is becoming increasingly important to find effective alternative ways of treating individuals with mental health needs to effectively help mitigate this danger (Walker et al., 2025).

Recent emerging technologies in the form of digital therapeutics (DTx) and artificially intelligent (AI) therapeutic-oriented software have been explored and marked as showing potential (Pham et al., 2022). Among therapeutic oriented software, AI has been highlighted as capable of providing significant usefulness when integrated or paired with such software, with affective computing systems and their use in Digital Mental Health Interventions (DMHIs) showing great capability in diagnosing, assisting, and directly working on the treatment of people with mental health conditions in people by enhancing user engagement (Schlicher et al., 2025). Implementing positive computing applications with existing mental health architecture and DMHIs can contribute to an improved overall experience (Lee et al., 2019). These software applications and systems carry the potential to be used to enhance and even provide direct mental health treatment to people with mental health needs. Despite this, hardware limitations play a role in the overall effectiveness of treatment (Wu et al., 2024).

Narrative

Mental health conditions cause problems and challenges for people diagnosed and undiagnosed with mental health related conditions. An early 2014 study on developing mental health applications for youth, highlighted that approximately 20 percent of adolescents were found to be affected by mental health disorders in established nations (Kenny et al., 2014). In a data table provided in a meta-analysis study conducted on deaths in people with mental health conditions, eight million deaths were attributed to mental disorders in 2012, and it was found that individuals with mental health conditions face an increasingly high level of death in terms of cause of deaths (Walker et al., 2025).

Methods

To gather relevant information for the research topic, a structured literature search approach was used. The primary and only search engine utilized for the research conducted was One Search. Only recent peer reviewed articles spanning from 2014 to 2026 with titles relevant to the topic were gathered and used as references for gathering and compiling useful information depending on relevancy found in information screened from abstracts that contained information applicable to the topic. Search queries consisted of "Mental Health" AND "Software" OR "Digital Therapy" AND "Mental Health," and these searches were narrowed down by filtering based on being a peer reviewed journal. Articles were primarily analyzed and filtered out by using a systematic review throughout.

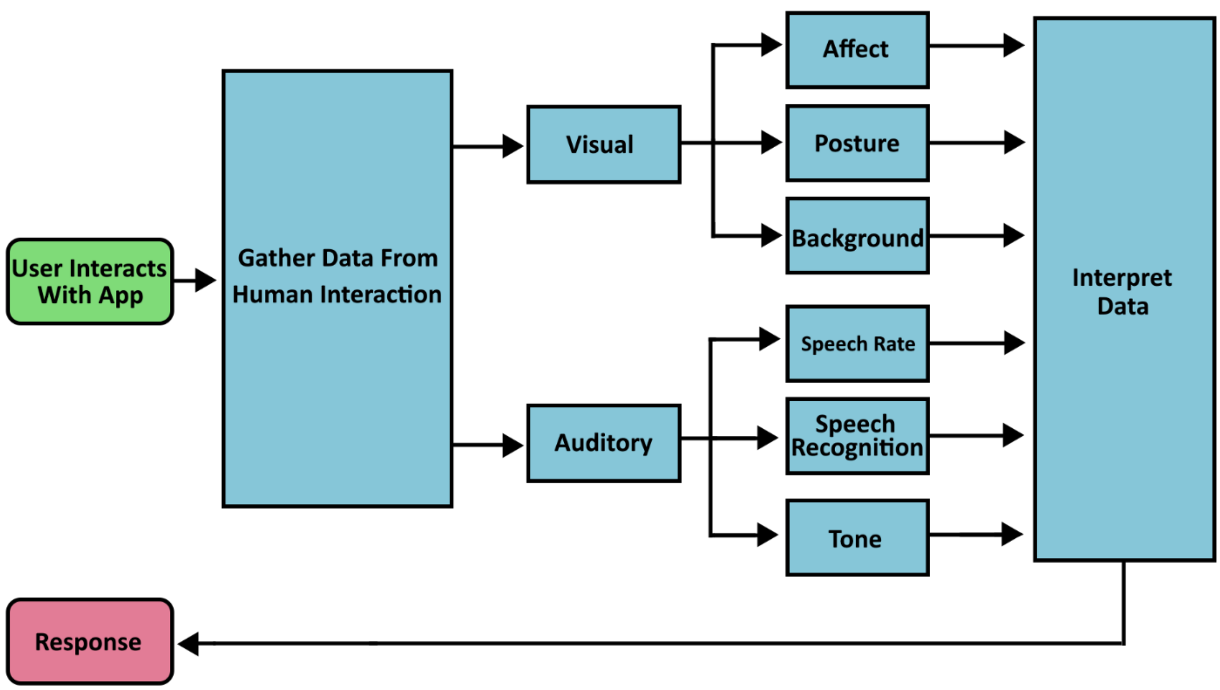
Results

DMHIs have been found to provide direct treatment to individuals by providing personalized treatment plans to individuals and have also been confirmed as effective in treating or assisting with the treatment of people with mental health conditions through the use of digital Cognitive Behavioral Health Therapy (CBT) (Lehtimaki et al., 2020). When such digital interventions are paired with an adaptive underlying architecture, the effectiveness of the systems can be further improved (Sundaram et al., 2024). DMHIs have been found to assist and even provide direct treatment to individuals facing mental health conditions, with results demonstrating a reduction of stress, depression, and anxiety in individuals (Schlicher et al., 2025). Digital interventions in the prevention and treatment of mental health disorders are proving to be increasingly cost effective; and DMHIs in the form of chatbots, internet therapy, virtual reality (VR), and augmented reality (AR); have shown positive results in directly treating people by reducing symptoms of mental health conditions or emotions (Lochner et al., 2025).

Recent studies on the implementation of digital therapeutic oriented software and its importance in finding alternatives for treating the increasing population of people with mental health needs have been conducted. A 2015 study on the use of digital therapeutic software in treating PTSD through the use of AR and VR found that symptoms of PTSD were reduced with consistent use of digital exposure therapy and CBT (Wu et al., 2024). Positive computing models that collect data to provide mental health treatment can utilize multiple wearable devices to improve the delivery of direct and effective interventions and some benefits of making use of positive computing are detecting the behaviors of humans that may be of distress and providing and tracking promptly responses to such behaviors to assist with measuring true effectiveness of such treatment (Lee et al., 2019). Despite this, hardware limitations in user comfort, reliability, and accuracy of wearable digital mental health devices can hinder the overall productivity of intended results (Wu et al., 2024); however, technological performance of DMHIs continues to increase (Schlicher et al., 2025).

When positive computing systems are paired with wearable technology, positive computing systems can provide appropriate responses to a user through mood or behavioral detectors that may utilize auditory and visual information (Lee et al., 2019). Similarly, a 2023 study on the use of AI chatbots to assist individuals with coping with the loss of a loved one highlighted that exposure therapy was a crucial part of successfully helping people cope with the loss (Jimenez-Alonso & Bresco de Luna, 2023). Likewise, affective computing systems show potential in increasing user engagement by providing insightful and useful responses to initials depending on their emotional state through the use of AI (Schlicher et al., 2025). These systems can be paired with a digital AI mental health software system framework known as the TEQUILA, or the trust, evidence, quality, usability, interest, liability, and accreditation framework to provide a generally responsible, efficient, and effective treatment (Lochner et al., 2025). Despite this, effectiveness of digital therapy can be unclear given the limitation of AI therapists in providing human-like interactions (Wu et al, 2024). Figure 1 illustrates an affective digital software system capable of gathering visual and auditory data to interpret a user's emotions and providing an appropriate response.

Figure 1. A generalized graphical overview of an affective computing system.



Note. Adapted from "Emotionally adaptive support: a narrative review of affective computing for mental health," by M. Schlicher et al., 2025, *Frontiers in Digital Health*, *7*, p. 7. Copyright 2025 by Schlicher, Li, Murthy, Sun and Schuller.

Discussion

Rising mental health conditions and rising death rates of individuals with mental health conditions highlight a need for alternative ways of providing mental health care. Digital therapeutics are newer regulated forms of software that are approved to provide direct and federally compliant treatment. Utilizing adaptive data-driven architectures alongside therapeutic oriented software can help provide efficient and effective treatment of people with mental health conditions by having a machine learning application capable of producing and providing a personalized treatment plan (Sundaram et al., 2024). Despite this, human accreditation, intervention, and oversight of these software systems during a user's interactions with digital therapeutics, especially AI-integrated treatment ones, is essential to preventing improper results (Lochner et al., 2024).

Primary and dominant costs and concerns of these software systems are privacy, ethical, security, and moral oriented. AI chatbots, despite showing benefits in assisting with managing a user's emotions, carry concerns of the security of the protected health information being transmitted (PHI), and its compliance with the Health Information Accountability and Portability Act (HIPAA) (Pham et al., 2022). For AI grief bots, ethical concerns regarding the capacity that grief bots can integrally represent a deceased person's memory (Jimenez-Alonso & Bresco de Luna, 2023). For DMHIs, data security, privacy, and improper treatment results should be taken into heavy consideration (Lochner et al., 2025). Affective mental health computing systems, in the commercial sense, raise moral concerns in that they can carry the downside of being prioritized to earn capital by keeping users attached longer rather than providing and promoting fast recovery (Schlicher et al., 2025).

Conclusion

Digital therapeutic-oriented software applications show incredible potential in directly assisting with the treatment of mental health individuals. By taking into account privacy, ethical, security, and moral concerns when developing, implementing, and employing such systems, the positives that arise in the form of successful results suggests that digital therapeutic oriented software can be effective in assisting with and even directly working on providing treatment some mental health conditions in people. Although direct treatment plans are not taken into full consideration and instead generalized throughout, a wide range employment of this software in the general sense may help reach and treat a higher number of individuals with mental health needs. Effective treatment and benefits of employing a mix of these digital therapeutics can be seen subtly, showing that these systems as a whole can be effective in treating individuals with mental health needs to a certain extent than not.

Primary concerns of implementing therapy-oriented software arise in the form of hardware limitations in the sense of efficient responses; privacy in terms of data being collected and potentially stored; ethicality in the sense of creating systems that can cause people to lose jobs; morality in the sense of employing systems that have the potential to produce incorrect outputs and lead to improper treatment of individuals; and security in the sense of potential data mishandling or leaks. Cost effectiveness of developing and employing therapeutic software can hinder effectiveness by making the software more difficult to create. Hardware limitations were not taken into full consideration throughout the study; however, these can and have been shown to carry the potential to hinder prompt and efficient interventions in digital systems running on lower end hardware. It is imperative to take these considerations into perspective when conducting further research delving into the effectiveness of digital therapeutics and if it is truly worth creating an app based on those perspectives.

References

Jiménez-Alonso, B., & Brescó de Luna, I. (2023). Griefbots. A New Way of Communicating With The Dead? *Integrative Psychological and Behavioral Science*, *57*, 466-481. https://doi.org/10.1007/s12124-022-09679-3

Kenny, R., Dooley, B., & Fitzgerald, A. (2014). Developing mental health mobile apps: Exploring adolescents' perspectives. *Health Informatics Journal*, *22*(2), 265–275. https://doi.org/10.1177/1460458214555041

Lee, U., Han, K., Cho, H., Chung, K.-M., Hong, H., Lee, S.-J., Noh, Y., Park, S., & Carroll, J. M. (2019). Intelligent positive computing with mobile, wearable, and IoT devices: Literature review and research directions. *Ad Hoc Networks*, *83*, 8–24. https://doi.org/10.1016/j.adhoc.2018.08.021

Lehtimaki, S., Martic, J., Wahl, B., Foster, K. T., & Schwalbe, N. (2020). Evidence on digital mental health interventions for adolescents and young people: A systematic overview. *JMIR Mental Health*, *8*(4), e25847. https://doi.org/10.2196/25847

Löchner, J., Carlbring, P., Schuller, B., Torous, J., & Sander, L. (2025). Digital interventions in mental health: An overview and future perspectives. *Internet Interventions*, *40*(100824), 100824. https://doi.org/10.1016/j.invent.2025.100824

Pham, K. T., Nabizadeh, A., & Selek, S. (2022). Artificial Intelligence and Chatbots in Psychiatry. *Psychiatric Quarterly*, *93*(1), 249–253. https://doi.org/10.1007/s11126-022-09973-8

Schlicher, M., Li, Y., Murthy, K., Sun, Q., & Schuller, B. W. (2025). Emotionally adaptive support: a narrative review of affective computing for mental health. *Frontiers in Digital Health*, *7*. https://doi.org/10.3389/fdgth.2025.1657031

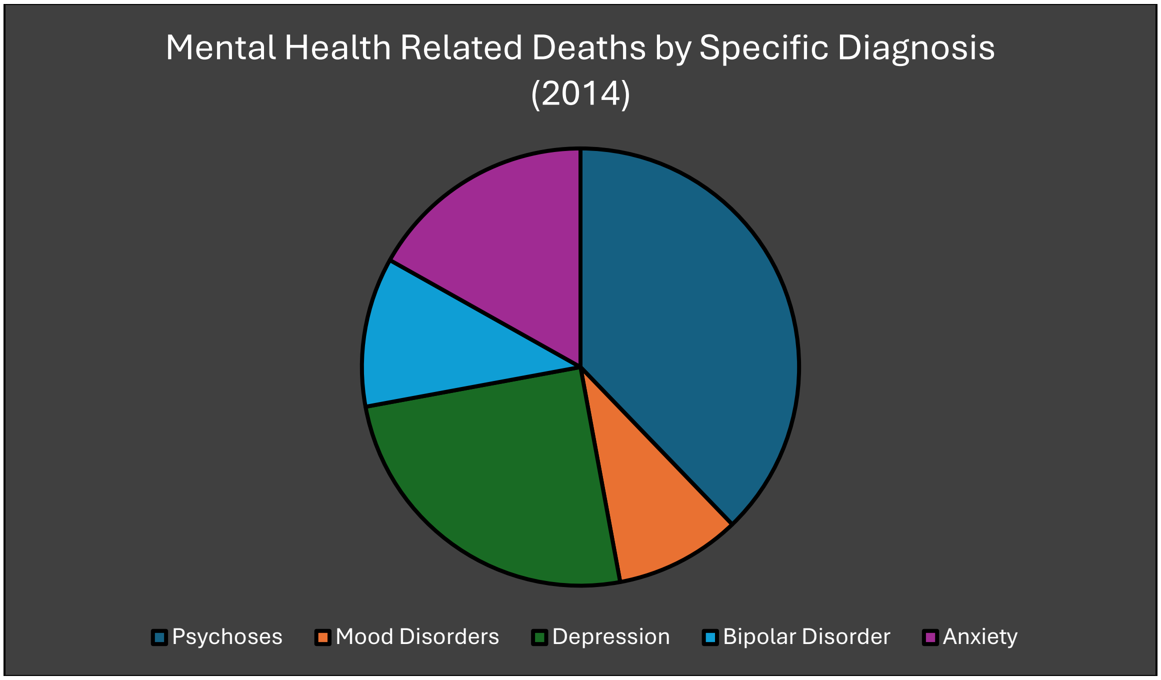
Sundaram, A., Subramaniam, H., Ab Hamid, S. H., & Mohamad Nor, A. (2024). An adaptive data-driven architecture for mental health care applications. *PeerJ*, *12*. https://doi.org/10.7717/peerj.17133

Wu, J.-Y., Tsai, Y.-Y., Chen, Y.-J., Hsiao, F.-C., Hsu, C.-H., Lin, Y.-F., & Liao, L.-D. (2024). Digital transformation of mental health therapy by integrating digitalized cognitive behavioral therapy and eye movement desensitization and reprocessing. *Medical & Biological Engineering & Computing*, *63*, 339–354. https://doi.org/10.1007/s11517-024-03209-6

Walker, E. R., McGee, R. E., & Druss, B. G. (2015). Mortality in Mental Disorders and Global Disease Burden Implications. *JAMA Psychiatry*, *72*(4), 334-341. https://doi.org/10.1001/jamapsychiatry.2014.2502

Appendix

Mental Health Related Deaths in 2014



Source: Data from "Mortality in Mental Disorders and Global Disease Burden Implications," by E. R. Walker et al., 2025, *JAMA Psychiatry*, *72*(4), p. 338. https://doi.org/10.1001/jamapsychiatry.2014.2502.